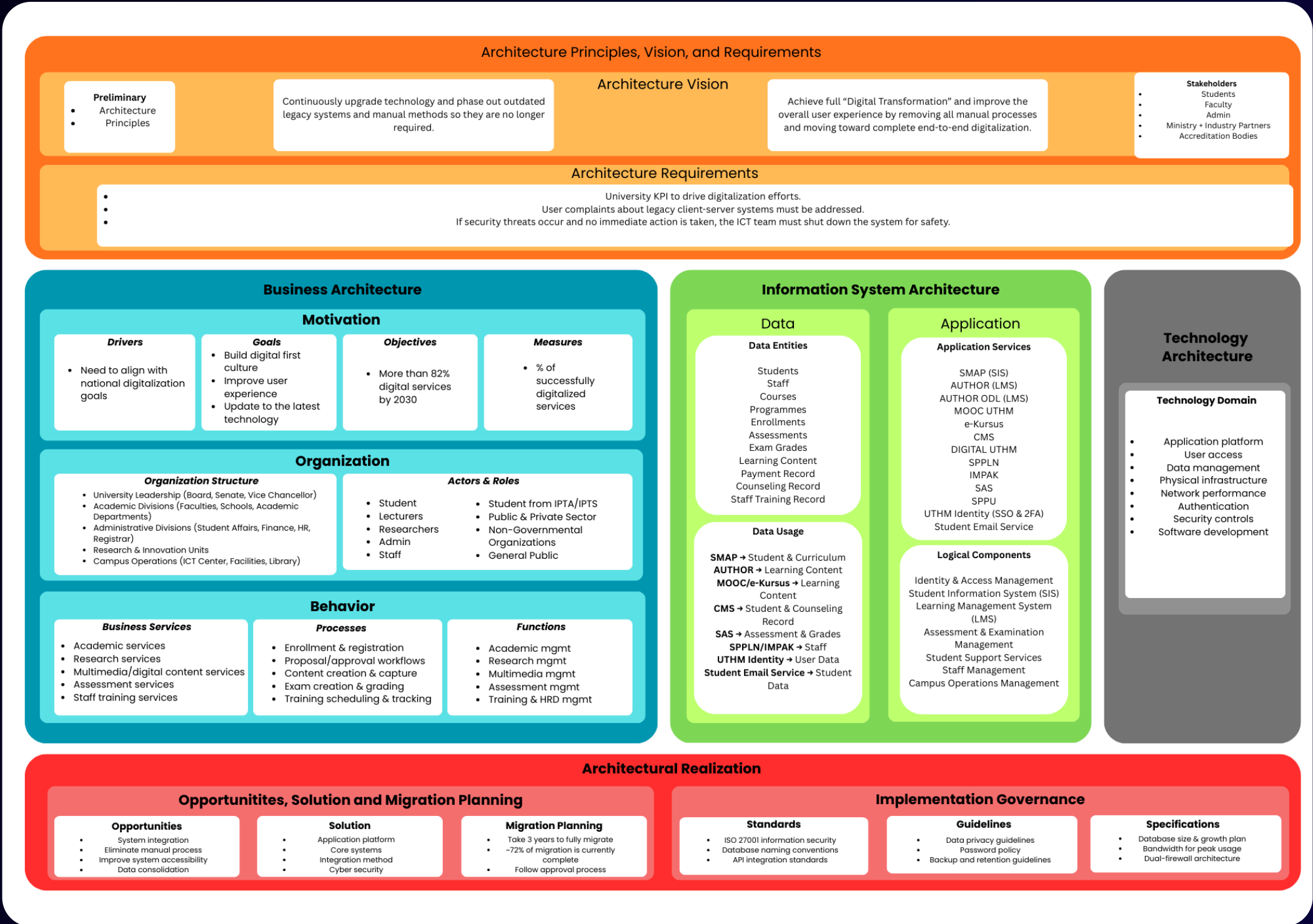


UTHM Enterprise Architecture

Prepared by SysDataverse



AS-IS Architecture



Architecture Principles, Vision and Requirements

Principles

- Replace outdated systems
- Reduce manual work

Vision

- Achieve full digital transformation
- Improve user experience
- Ensure all services are connected

Requirements

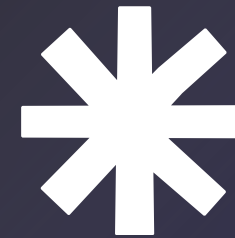
- Collect and consider users' feedback
- Maintain high system security
- ICT team must respond immediately to any security issues

Manual & Paper Based Process

- PKU relies heavily on manual documentation
- Slow patient registration
- Difficulty in tracking medical histories

AS-IS Problems

Current State



Analytics & Business Intelligence Issues

- UTHM has massive amounts of data
- Data trapped in separate systems
- No Real-Time view for university management

To-Be Business Architecture



STRATEGY

- Driver

External pressure for 'Always Online' systems. Need for real-time analytics to replace manual reporting.

- Goals

Establish a Hybrid Architecture: Combining On-Premise security with Cloud scalability

- Objective

100% Paperless & Cashless. Zero physical forms for administrative tasks.

- Measures

99.99% availability for LMS

ORGANIZATION

- New hybrid unit

Hybrid Infrastructure & Integration Unit to bridge legacy with cloud.

- New roles

Integration Specialists & Data Center Ops Team.

SERVICES

- Personalized Learning

AI driven content adaptive to student needs.

- Smart Campus

IoT-enabled Parking & Library access.

PROCESS & FUNCTIONS

- Automated Enrollment

Instant regulation without human approval delays.

- AI-Protocol Exams

Autopiloted online assessment ensuring integrity.

- Hybrid Operation Management

Single-pane view of IT health (Cloud + On-Premise).

To-Be Application Architecture *

SERVICE

- Learning Platforms
- Administrative Platforms
- Enterprise Services

LOGICAL COMPONENT

- **Identity & Access Management:** Authentication and user rights.
- **Student Information System (SIS):** This is the database of student lifecycle.
- **Learning Management (LMS):** The teaching, learning, and collaboration interface.
- **Assessment and examination:** Intensity course modules: grading and exam integrity.
- **Campus Operations:** Systems: Research and day-to-day logistical systems.
- **Health Management:** A safe university medical service system.

DATA ENTITIES

- Essential Academic Branches: Students, personnel, courses, and programs.
- Academic history: classifications, examinations, and grades.
- Operational Assets: Learning material and payment history.
- **Sensitive Records:** Special training and medical history records.

To-Be Data Architecture



DATA USAGE

- LMS Integration
- Administrative Management
- Security & Identity
- **Strategic Intelligence:** Business intelligence (BI) tools combine data to provide a management dashboard.
- **Governance:** The medical data access is highly divided and limited to the privacy policies.

To-Be Technology Architecture



Centralized User Access

- One system manages all user accounts
- Standard access rules, easy onboarding/offboarding

Managed Data Services

- Central data storage
- Regular backups, consistent data, easy reporting

Hybrid Infrastructure

- Mix of on-premise + cloud
- Sensitive data on-premise, scalable apps on cloud

Optimized Network

- Faster internet & strong WiFi
- Network segmentation for safety
- Stable online learning

Fully Integrated SSO

- One login for all systems (SIS, LMS, HR, Library, Finance, PKU)
- Central identity management
- Ready for MFA

Enhanced Security

- Firewalls, monitoring, encryption
- PDPA compliance
- Zero Trust approach
- Regular security audits

GAP ANALYSIS

CURRENT:

- Mostly digital process, but 20–30% remain manual.
- Siloed legacy applications causing performance issues.
- Infrastructure mostly on-premise lead to bottlenecks during peak usage.

TO-BE:

- Fully digital and paperless.
- Integrated systems with Master Data Management.
- Hybrid cloud infrastructure to handle high-traffic applications

Gap Analysis and Roadmap

ROADMAP *

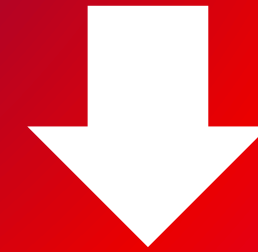
- **Short-term:** rationalize systems & consolidate processes into digital modules.
- **Medium-term:** deploy a hybrid cloud, strengthen connectivity & enhance security.
- **Long-term:** implement smart campus IoT services and automate workflows.

Conclusion

Values

- Efficient & Paperless Operations
- Integrated & Secure Systems
- Better Decisions & User Experience

Manual processes, siloed systems, and limited analytics visibility.



The proposed TO-BE Enterprise Architecture introduces a **hybrid cloud model** that balances security, scalability, and performance.

Q&A Session